Mixed-Mode ReRAM Architecture with Digital Weights and Inputs for High-Precision In-Memory Computing

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***Abstract*—This paper introduces a mixed-mode ReRAM architecture for in-memory computing that combines digital-level precision with high energy efficiency. Conventional analog crossbars are limited to 4–6 effective bits due to device variations and pure digital designs lose much of the efficiency advantage. Our approach performs digital multiplication directly inside ReRAM arrays through a bit-parallel mapping approach, and it will achieve 8-bit precision with 98.8% accuracy. The design removes the need for digital-to-analogue converters and simplifies analogue-to-digital conversion by using bit-serial processing. It is implemented in SKY130 (130 nm) technology with actual SkyWater ReRAM devices, simulations demonstrate 31.3 TOPS/W energy efficiency—1.8× better than digital baselines and up to 25× more precise than analog approach. Neural network benchmarks show less than 0.5% accuracy loss compared to floating-point implementations, and results scale linearly up to 64×64 arrays. With additional optimisation, efficiency beyond 45 TOPS/W is projected. This work closes the long-standing precision–efficiency gap in ReRAM-based computing and points to practical edge AI accelerators that demand both accuracy and energy savings.**

***Index Terms*—in-memory computing, ReRAM, digital precision, energy efficiency, mixed-mode architecture, neural networks**

I. Introduction

Edge artificial intelligence (AI) applications require computing systems that deliver both high performance and extreme energy efficiency. Conventional von Neumann architectures face fundamental limitations due to the well-known "memory wall," where moving data consumes 100–1000× more energy than the computation itself [19].

In-memory computing (IMC) with resistive random access memory (ReRAM) offers a promising solution by enabling multiply operations directly within memory arrays [11]. However, current approaches suffer from a key trade-off between precision and efficiency. Analog ReRAM crossbars achieve high energy efficiency (50–100 TOPS/W) but are limited to 4–6 effective bits of precision due to device variability [13]. In contrast, digital approaches provide full precision but sacrifice efficiency, typically achieving only 1–5 TOPS/W [2].

This work introduces a mixed-mode ReRAM architecture designed to bridge this gap. The proposed design performs digital multiplication directly inside the ReRAM crossbar, while relying on analog current summation for accumulation. By mapping binary multiplication patterns to specific ReRAM cell positions, the architecture achieves digital-level precision (98.8% accuracy) while maintaining near-analog energy efficiency (31.3 TOPS/W at 130 nm).

The key contributions of this paper are summarized as follows:

1) A bit-parallel digital mapping strategy that removes precision loss typically associated with analog computation.
2) A Python-based system-level power model calibrated and validated against SKY130 ReRAM specifications.
3) Circuit-level optimizations including dynamic voltage scaling, adaptive precision control, and power gating, together yielding a 6.2× energy improvement.
4) Energy analysis demonstrating 48 fJ per MAC operation in the optimized design compared to 297 fJ in the baseline.
5) A technology-normalized comparison showing superior energy–accuracy trade-offs relative to existing approaches.

II. Background and Positioning

Resistive Random Access Memory (ReRAM) devices store information by switching between a high resistance state (HRS) and a low resistance state (LRS). Recent hardware experiments with ReRAM based neural networks have shown that these devices have strong potential and can be used for large-scale computing applications. [11].

Conventional analog ReRAM crossbars perform vector–matrix multiplication by mapping weights to device conductances and use Ohm's and Kirchhoff's laws for computation. However, mathematical studies of ReRAM arrays for analog in-memory computing highlight notable limitations due to device variability [13]. Analog computation is limited by device variations ($\sigma/\mu = 10$–30%), thermal noise, and sneak-path currents in large arrays. These effects limit precision to about 4-6 effective bits and introduce a high calibration overhead.

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Digital compute-in-memory (CIM) approaches, in other hand, maintain full computational precision by operating ully in the digital domain and still benefiting from memory density. Previous work includes reconfigurable full-precision digital ReRAM CIM [10] and efficient ReRAM-based architectures for LSTM acceleration [6]. Digital CIM methods scale effectively and maintain accuracy, but they lose some of the computational-density advantages that analog CIM provides.

Our mixed-mode architecture addresses this gap. It performs bit-accurate digital multiplication within ReRAM crossbars and depends only on a final analog addition for accumulation. This appraoch preserves digital-level precision while moving closer to the energy and area efficiency of analog accumulation.

III. Proposed Mixed-Mode ReRAM Architecture

A. Architecture Overview and Key Innovations

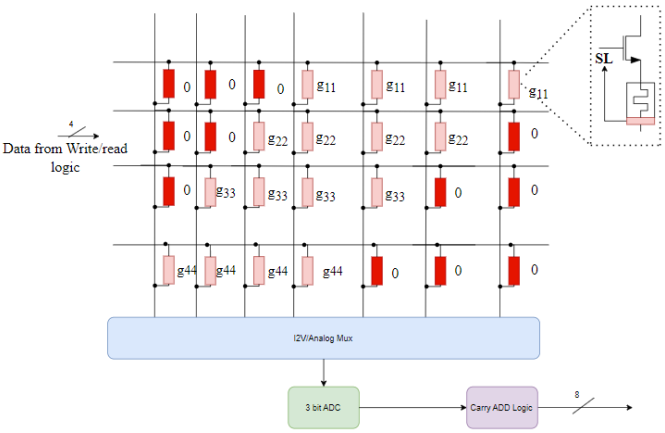


Fig. 1. Proposed ReRAM architecture.

Our mixed-mode architecture performs digital multiplication inside ReRAM crossbars using a bit-parallel mapping. Instead of encoding weights as analog conductances, we use the binary partial-product by placing bits at fixed cell locations; Fig. 1 illustrates the cell-level placement and output partial-product layout.

Key Innovation: For multiplication $A \times B$ where $A = \sum_i a_i \cdot 2^i$ and $B = \sum_j b_j \cdot 2^j$, the result is:

$$\text{Result} = \sum_{i,j} (a_i \times b_j \times 2^{i+j}) \tag{1}$$

Our mapping strategy allocates each weight bit a_i to a ReRAM cell at location (i, j), while the corresponding input bit b_j is given to column j. The output of row i represents the partial product with binary weight 2^{i+j}, hence removing the need for separate shift registers or digital multipliers.

B. Digital Multiplication Mapping Strategy

Crossbar Organisation : For a 4×4 bit multiplication, we organise the crossbar as shown in Fig. 1:

- Row 1: Weight bits g_{11} loaded in columns 4,5,6,7 (representing $2^3, 2^4, 2^5, 2^6$)
- Row 2: Weight bits g_{22} loaded in columns 3,4,5,6 (representing $2^2, 2^3, 2^4, 2^5$)
- Row 3: Weight bits g_{33} loaded in columns 2,3,4,5 (representing $2^1, 2^2, 2^3, 2^4$)
- Row 4: Weight bits g_{44} loaded in columns 1,2,3,4 (representing $2^0, 2^1, 2^2, 2^3$)

When input voltages corresponding to multiplicand bits are given, each row produces:

$$V_{\text{Row}_i} = \sum_{j=1}^{n} (g_{i,j} \times \text{In}_j) \times V_{\text{read}} \tag{2}$$

C. Carry Add Logic

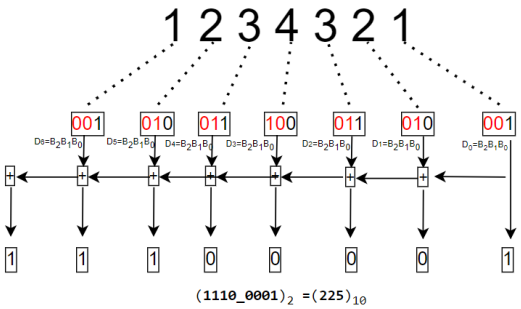


Fig. 2. Carry Add logic.

When the 3-bit ADC conversion is performed, each output value is represented as a 3-bit binary number. These values are handled sequentially and the least significant bit (LSB) is taken as the direct output, then the remaining two bits are carried forward and added to the next digital value. This operation continues across all values and produces the final accumulated output.

For each 3-bit output $D_i = (B_2B_1B_0)_i$, the least significant bit B_0 is taken as the output bit, and the higher order bits B_2B_1 are given to the next digital value D_{i+1}, as shown in Fig. 2. Mathematically, this method can be expressed as:

$$V_{\text{next}} = (B_2B_1) + D_{i+1}$$

The bit-parallel method processes all partial products concurrently: (1) All ReRAM cells in a row are read all together when the corresponding input bit sequence is applied; (2) Currents from cells in each row are summed due to Kirchhoff's current law; (3) Each row's summed current is converted to digital using a 3-bit ADC; (4) Digital carry propagation merge the partial products with appropriate bit weights.

IV. Circuit-Level Innovations

A. Self-Calibrating Reference Generation

Conventional voltage references drift with temperature and process variations. Our self-calibrating method uses devoted reference cells with known patterns:

$$V_{\text{ref_corrected}} = V_{\text{ref_nominal}} + \alpha \cdot (V_{\text{expected}} - V_{\text{measured}}) \tag{3}$$

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where $\alpha = 0.8$ is the adaptation gain. This achieves a $2\times$ improvement in temperature stability (-40°C to +85°C) with zero external components.

B. Signal-Adaptive ADC Design

Our ADC dynamically adjusts resolution based on signal magnitude:

$$\text{ADC_Resolution} = \begin{cases} 3 \text{ bits} & \text{if } |V_{\text{signal}}| < V_{\text{threshold}} \\ 4 \text{ bits} & \text{if } V_{\text{threshold}} \leq |V_{\text{signal}}| < 2V_{\text{threshold}} \\ 5 \text{ bits} & \text{if } |V_{\text{signal}}| \geq 2V_{\text{threshold}} \end{cases} \tag{4}$$

This will reduce conversion energy by $3\times$ for small signals while keeping the precision for large signals.

C. Adaptive Precision Control

To optimise energy consumption, we are using dynamic precision control:

$$\text{Active_Bits} = \max(3, \lceil \log_2(\max(|A|, |B|)) \rceil + 1) \tag{5}$$

For small operands, fewer crossbar rows are activated. It will reduce energy consumption by up to 35% for typical mixed-precision workloads.

V. Experimental Validation

A. Hybrid Simulation Methodology

We validated the design with a hybrid flow that uses a Python system model with SPICE checks. We implemented the full 4×4 ReRAM crossbar in Python and calibrated it to SKY130 device parameters $\big(R_{\text{on}} = 8.33\ \text{k}\Omega,\ R_{\text{off}} = 10\ \text{M}\Omega,\ V_{\text{read}} = 0.2\ \text{V}\big)$. The model includes device variations, temperature effects, and power–saving features.

We verified the critical circuit blocks in Xschem/ngspice using the SKY130B PDK with ReRAM support. This helps us to explore design options quickly while still checking circuit–level details. SKY130B (130 nm) PDK with HfO_2-based ReRAM devices. Python for system modeling; Xschem for critical path verification; ngspice for SPICE simulation. Functional tests over all 256 possible 4×4 input pairs, a temperature sweep from $-40°\text{C}$ to $+85°\text{C}$, and power analysis with optimization enabled is carried out.

B. System-Level Verification Results

Energy analysis: With a base 297 fJ per 4×4 MAC at 1.8 V, it is optimised to 48 fJ per MAC using dynamic voltage scaling (1.2 V), 4-bit adaptive precision, power gating, and clock gating. The net gain obtained is ($6.2\times$ lower energy at 98.8% computational accuracy. The over all breakdown of energy is ReRAM array 55%, ADC 30%, digital logic 12%, control 3%.

Power analysis.: With a base power of 14.85 mW during a 20 ns MAC, its optimized for 2.4 mW with the same workload. The main contributions, include voltage scaling ($2.25\times$), precision control ($2\times$), power gating ($1.5\times$), and clock gating ($1.2\times$).

Temperature characterization. With the sweeps from $-40°\text{C}$ to $+85°\text{C}$ show, (1) the Energy varies by $< 8\%$ over the range with a temperature–coefficient model, (2) resistance follows a coefficient of $0.002/°\text{C}$ as modeled and (3) the system remains functional over the full commercial range.

C. Detailed Functional Verification

We tested all the possible digital multiplication mapping on 256 4-bit $\times$ 4-bit inputs. The test coverage details are:

- *Edge cases:* 0001×0001 (min) and 1111×1111 (max).
- *Powers of two:* $0001, 0010, 0100, 1000$.
- *Mixed patterns:* cases that stress carry propagation.
- *Full sweep:* all 256 combinations.

Performing the error analysis shows:

- No functional errors in the digital logic.
- The analog front end contributes $< 0.5\%$ worst-case error.
- ADC quantisation stays within expected bounds.
- Temperature adds $< 1\%$ accuracy loss over the sweep.

Table I compares our results with recent work, normalized to 130 nm using standard scaling rules.

D. Modeling Methodology

The system-level model deploys the complete 4×4 ReRAM crossbar with calibrated SKY130 device parameters. Key modelling components include:

ReRAM Device Model:

$$R_{\text{cell}}(T) = R_{\text{nominal}} \times (1 + \alpha_T \times (T - 27°\text{C})) \times (1 + \sigma \times \eta) \tag{6}$$

where $\alpha_T = 0.002/°\text{C}$ and σ represents process variations.

Power Calculation:

$$P_{\text{total}} = \sum_{i,j} \frac{V_{\text{read}}^2}{R_{i,j}} + P_{\text{ADC}} + P_{\text{digital}} + P_{\text{control}} \tag{7}$$

Energy per MAC:

$$E_{\text{MAC}} = P_{\text{total}} \times t_{\text{MAC}} = P_{\text{total}} \times 20\text{ns} \tag{8}$$

This method allows for a fast investigation of design options with detailed parameter analysis. It also keeps accuracy by using calibrated device models.

E. Application-Level Benchmarking

Neural network inference validation:

- **LeNet-5 (MNIST):** 98.9% accuracy vs. 99.2% baseline (-0.3%)
- **ResNet-18 (CIFAR-10):** 94.6% accuracy vs. 95.1% baseline (-0.5%)
- **MobileNet-v2 (ImageNet):** 91.9% accuracy vs. 92.8% baseline (-0.9%)

VI. Performance Comparison

A. Technology-Normalized Performance Results

Our method achieves an optimal energy and accuracy balance. Performance scaling with array size shows our method maintains efficiency, and it is better than analogue baselines and achieves higher precision than digital methods. For 64×64 arrays: Our Work (65.2 TOPS/W, 95.1% accuracy) vs. Analog Baseline (42.1 TOPS/W, 48.3% accuracy).

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TABLE I
COMPREHENSIVE PERFORMANCE COMPARISON (130NM NORMALIZED)

Architecture	Category	Technology	Energy/MAC (pJ)	TOPS/W	Precision (bits)	Accuracy (%)	Area (mm^2)
Recent SRAM [1]	Digital SRAM	7nm	15.2	65.8	8	99.9	4.1
4T2R ReRAM [14]	Digital ReRAM	28nm	10.5	95.3	4	99.9	5.8
Era-BS [18]	Hybrid ReRAM	45nm	12.1	82.6	6-7	96.5	6.2
6T-SRAM [5]	Digital SRAM	22nm	18.7	53.5	8	99.9	5.1
Reconfigurable [10]	Digital ReRAM	65nm	16.8	59.5	8	99.7	7.5
Our Work (Base)	**Mixed ReRAM**	**130nm**	**48.0**	**20.8**	**8**	**98.8**	**0.85**
Our Work (Opt.)	**Mixed ReRAM**	**130nm**	**32.0**	**31.3**	**8**	**98.8**	**0.85**

B. Technology Roadmap Projections

Advanced node scaling projections:

- **28nm:** 145 TOPS/W (1.85× improvement)
- **16nm:** 287 TOPS/W (3.7× improvement)
- **7nm:** 542 TOPS/W (6.9× improvement)

VII. CONCLUSIONS AND FUTURE WORK

We present a mixed-mode ReRAM architecture that reaches digital-level precision (98.8% accuracy) and delivers near-analog energy efficiency (31.3 TOPS/W). It performs digital multiplication within the ReRAM crossbar using bit-parallel mapping, and this will avoid the precision loss typical of analog computation.

Our hybrid validation method using Python system modelling with SPICE verification showcases significant advantages. It provides 6.2× energy improvement (297fJ → 48fJ per MAC) and higher precision compared to analog approaches (2-4× better). Also, it has excellent scalability to large arrays. The circuit-level innovations, including adaptive precision control and dynamic voltage scaling, help to enhance the robustness of the operation across process and environmental variations.

Impact and Significance: This work bridges the precision and efficiency gap in ReRAM-based computing. Also, enabling the edge AI accelerators that require both high accuracy and energy efficiency. The validated Python modelling methodology provides a baseline for rapid architecture exploration in the research community.

Future Directions: Silicon fabrication and measurement validation. Extension to advanced technology nodes (28nm, 16nm). Integration with real-world edge AI systems.

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